ME 351: Feedback Control Systems

Fall 2026

The Cooper Union • Albert Nerken School of Engineering

Instructor	Dr.-Ing. Dirk M. Luchtenburg	**Office**	41 Cooper Square, Room 719
Email	dirk.luchtenburg@cooper.edu	**Prerequisite**	Systems Engineering (ME 251 / ESC 251)
Class meetings	Monday 4:00–4:50 PM, 41CS 505	Thursday 3:00–4:50 PM, 41CS 504	
Office hours	*To be announced; additional meetings available by appointment.*		

Course at a Glance

Feedback control is the engineering discipline of influencing dynamical systems so that they behave in desired ways despite disturbances, uncertainty, and imperfect measurements, within the limits imposed by physics, sensing, and actuation.

The course is organized around four recurring engineering questions:

Representation $\longrightarrow$ **Behavior** $\longrightarrow$ **Objective** $\longrightarrow$ **Decision**

Representation	How should we model the physical system for the question at hand?
Behavior	What does the model tell us about motion, stability, response, sensitivity, and robustness?
Objective	What behavior do we actually want, and how should performance be quantified?
Decision	What control action or controller structure should we choose, and how do we know it will work on the physical system?

> *We will use mathematics, computation, and experiments to answer engineering questions: how a system behaves, what limits that behavior, what performance is physically achievable, and what control action is appropriate.*

Course Description and Purpose

Official catalog description. Modeling and representation of dynamic physical systems: transfer functions, block diagrams, state equations, and transient response. Principles of feedback control and linear analysis including root locus and frequency response methods. Practical applications and computer simulations using MATLAB. Discussions of ethics will be integrated into the curriculum.

Course purpose. This course develops the principles and tools needed to analyze and design feedback control systems. The emphasis is not only on using standard control methods, but on understanding why those methods work, what assumptions they depend on, and how an engineer chooses among them.

We will repeatedly move between physical models, mathematical representations, computation, and engineering interpretation. Linear control theory is the main analytical framework, but nonlinear physical models will be used where useful to understand operating points, linearization, model validity, and the practical range of linear controllers.

Professional responsibility is integrated with the technical material. Control engineers often work on systems whose safety depends on recognizing instability, saturation, uncertainty, and fundamental limits before those limits are encountered in operation. Technical competence, careful validation, and clear communication of risk are therefore treated as part of responsible engineering practice.

Learning Outcomes

By the end of the course, you should be able to:

Representation. Develop purposeful dynamic models from physical principles; distinguish states, inputs, outputs, disturbances, and parameters; write nonlinear state-space models; identify operating points; linearize locally; and move between state-space, transfer-function, and block-diagram representations.

Behavior. Analyze natural and forced response, poles and modes, transient behavior, stability, frequency response, sensitivity, robustness, and the effects of feedback on disturbances and uncertainty.

Objective. Translate engineering goals into meaningful specifications involving tracking, regulation, disturbance rejection, noise attenuation, transient response, robustness, and control effort.

Decision / Design. Design and assess controllers using proportional, integral, derivative, root-locus, frequency-domain, loop-shaping, state-feedback, observer, and introductory optimal-control methods.

Computation. Use Python and, where useful, MATLAB to simulate dynamic systems, compare linear and nonlinear models, analyze control systems, design controllers, and verify that numerical results are physically credible.

Engineering Judgment. State modeling assumptions; estimate physical scales before calculation; identify actuator, sensing, model, and *fundamental performance limitations*; validate simplified models; recognize when a design is operating too close to a physical or theoretical limit; and communicate the reasoning behind a control design.

Professional Responsibility. Recognize that technical competence is part of ethical engineering practice: identify risks and limitations, quantify them where possible, communicate them honestly, and recognize when professional responsibility requires redesign, restricted operation, additional testing, or escalation of a safety concern. A useful progression is

$$\boxed{\text{Competence} \to \text{Limits} \to \text{Risk} \to \text{Communication} \to \text{Action}}$$

Course Materials

Primary text. Karl J. Åström and Richard M. Murray, *Feedback Systems: An Introduction for Scientists and Engineers*, 2nd ed. A freely available electronic edition and supplemental materials are available at fbsbook.org.

Feedback Control Companion Notes. Course companion notes will be provided throughout the semester. They are not intended to replace a conventional textbook. Their purpose is to connect the mathematics to the engineering questions that motivate it and to make explicit the reasoning, assumptions, approximations, and computational checks that experienced control engineers use.

Supplemental references. Useful alternate treatments include:

- Randal W. Beard et al., *Introduction to Feedback Control Using Design Studies.*
- Franklin, Powell, and Emami-Naeini, *Feedback Control of Dynamic Systems.*
- Katsuhiko Ogata, *Modern Control Engineering.*
- Richard C. Dorf and Robert H. Bishop, *Modern Control Systems.*

Different texts use different notation, conventions, and points of emphasis; learning to translate between them is part of becoming technically fluent.

Software. Python will be used extensively for modeling, simulation, visualization, and control analysis. MATLAB may also be used where appropriate. Code should support engineering reasoning rather than replace it.

How the Course Will Work

The weekly rhythm will normally include a combination of:

- concept development and engineering examples;
- short analytical derivations;
- worked design or modeling problems;
- computational investigations in Python;
- interpretation of plots, simulations, and experimental behavior;
- homework and exam problems that require both calculation and engineering explanation.

Reading the companion notes and assigned textbook sections before and after class is strongly recommended. Control is cumulative: later topics depend heavily on modeling, dynamics, linear systems, and earlier feedback concepts.

Homework format and submission. Unless otherwise stated, homework will be prepared in LaTeX using the provided course template and submitted as a single PDF through Microsoft Teams. The goal is not typography for its own sake: a consistent technical format makes equations, assumptions, plots, and engineering reasoning easier to read, review, and discuss. Source files may also be requested when an assignment includes computation or reproducibility requirements.

Assessment

Fall 2026 weighting.

Problem sets, short quizzes, in-class work, and ethics reflection	20%
Computational investigations	15%
Two midterm examinations	35%
Final examination (cumulative)	30%
Total	100%

Examinations assess individual understanding. Homework and computational work are intended primarily for learning and practice; the expectations for collaboration will be stated on each assignment.

An engineering-ethics assignment will connect technical analysis to professional responsibility. The emphasis will be on competence, fundamental limitations, safety margins, honest communication of uncertainty and risk, and the engineer's responsibility to raise concerns when a design or operating practice is unsafe.

Letter-grade thresholds and exam dates will be announced before the first graded assessment. The instructor may lower, but will not raise, announced thresholds if adjustment is warranted by assessment difficulty.

Problem Solving and Submitted Work

A technically correct final number is not, by itself, a complete engineering solution. Unless a problem clearly requires otherwise, submitted work should make the following visible:

- the engineering question being answered;
- the variables and coordinate/sign conventions being used;
- important modeling assumptions;
- the governing equations or control relationships;
- enough algebra or computation to make the reasoning auditable;
- units and physically meaningful numerical values;
- a brief interpretation or sanity check when appropriate.

When computation is used, plots should have labeled axes and units, and code should be sufficiently organized that another engineer could reproduce the result.

Collaboration, Academic Integrity, and AI Tools

Collaboration

Control engineering benefits from discussion. Unless an assignment is explicitly designated as individual work, you are encouraged to discuss concepts, approaches, debugging strategies, and interpretations with classmates. Your submitted work must nevertheless reflect your own understanding and your own final presentation unless a team submission is explicitly requested.

When another person materially contributes to your solution, acknowledge that collaboration briefly.

Academic Integrity

Students are expected to follow The Cooper Union's Academic Standards and Regulations: Academic Standards and Regulations.

Examinations are individual work unless explicitly stated otherwise. Unauthorized copying, use of prior solution sets, or presentation of another person's work as your own is not permitted.

Use of AI Tools

Generative-AI tools may be useful for explanation, brainstorming, debugging, code review, and checking alternative approaches. They may also produce confidently incorrect mechanics, inconsistent notation, fabricated references, or code that does not represent the intended physical model.

Unless an assignment or examination states otherwise:

- AI tools may be used as a *learning aid*, not as a substitute for your own mechanics and engineering reasoning;
- you are responsible for understanding and being able to explain the equations, assumptions, claims, citations, and code you submit;
- substantive AI assistance should be disclosed briefly, including the tool and how it was used;
- you should expect to demonstrate independently, in quizzes, examinations, or brief oral discussion, the methods and reasoning used in submitted work;
- AI tools are not permitted on examinations unless explicitly authorized.

Communication, Attendance, and Deadlines

Communication. Course announcements, assignments, supporting files, and routine course communication will be distributed through Microsoft Teams. Email is appropriate for individual or sensitive matters. Please allow approximately one working day for a reply; questions sent immediately before a deadline may not receive a response in time.

Attendance and participation. Regular attendance is strongly expected. Class meetings will include derivations, discussion, computational demonstrations, and engineering reasoning that may not appear verbatim in the text or notes. If you must miss class, communicate when possible and take responsibility for determining what you missed.

Deadlines. Assignment deadlines and late-work rules will be stated clearly with each assignment. If significant circumstances interfere with your work, communicate as early as possible rather than after the deadline.

Accessibility and Learning Environment

Students with approved accommodations should provide the appropriate documentation through The Cooper Union's established process and contact the instructor as early as possible so that arrangements can be implemented effectively.

We aim to maintain a professional learning environment in which students can ask questions, make mistakes, challenge technical assumptions, and contribute different perspectives respectfully. If something is interfering with your ability to participate or learn effectively, please communicate with the instructor or the appropriate student-support office.

Tentative Course Roadmap

The schedule below describes the intended intellectual progression of the course. Exact pacing may change in response to class needs.

Representation → Behavior → Objective → Decision

Wk	Lens	Main questions and topics
1	Representation / Big Picture	**Thinking like a control engineer.** What is control? Dynamical systems; open loop, feedforward, and feedback; reference, output, error, disturbances, uncertainty, sensing, actuation, estimation, and physical constraints. Introduce the recurring engineering questions of Representation, Behavior, Objective, and Decision.
2	Representation	**From physical system to dynamic model.** States, inputs, outputs, parameters, and disturbances; first-principles modeling; equations of motion; nonlinear state-space form; equilibria and operating points; deviation variables and local linearization; model hierarchy and validity range; dimensional analysis and scaling; physical time scales; numerical integration and simulation.
3	Representation → Behavior	**From equations to system response.** Linear ordinary differential equations; transfer functions from differential equations; the Laplace transform as a computational tool; first- and second-order systems; poles and zeros; stability; time constants; step and impulse response; dominant poles; initial- and final-value reasoning.
4	Behavior	**What do poles tell us about dynamics?** Natural modes; damping; natural frequency; oscillation; resonance; overshoot; settling time; unstable growth; double-integrator behavior; relationship between pole location and transient response; preliminary root locus: how closed-loop poles move as feedback gain changes.
5	Objective → Decision	**What should feedback accomplish?** Tracking and regulation; steady-state error; feedforward plus feedback; proportional, integral, and derivative action; P, PI, PD, and PID structure; actuator saturation; integral windup; derivative filtering; practical controller choices.
6	Representation / Decision / Evaluate	**How do we tune a practical controller?** Simple low-order process models; integrating and first-order-plus-dead-time models; extracting gain, delay, and time constants from response data; model-based PI/PID tuning; IMC/SIMC philosophy; comparison with Ziegler–Nichols; anti-windup; uncertainty and actuator-effort tests.
7	Behavior / Evaluate	**How does a system behave across frequency?** Complex exponential and sinusoidal response; frequency response; magnitude and phase; Bode plots; effects of integrators, poles, and zeros; resonances; crossover frequencies; bandwidth; gain and phase margins; return to root locus; conceptual introduction to the Nyquist viewpoint.

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Wk	Lens	Main questions and topics
8	Decision / Evaluate	**How do we shape feedback—and what limits us?** Loop shaping; lead and lag compensation; crossover and bandwidth selection; sensitivity and complementary sensitivity; tracking, disturbance rejection, measurement noise, and model uncertainty; delay; right-half-plane poles and zeros; actuator and sensor limitations; feasible bandwidth; Bode sensitivity integral and the waterbed effect; fundamental limitations of feedback.
9	Representation → Behavior	**What is happening inside the system?** State-space representation; physical state selection; state-transition behavior; eigenvalues, eigenvectors, and modes; internal dynamics; realizations; relationship between transfer functions and state-space models; distinction between internal and input–output behavior.
10	Behavior → Decision	**Which internal dynamics can we control?** Controllability; actuator placement; state feedback; pole placement; reference tracking; prefilters; integral augmentation; practical control authority; distinguishing mathematical controllability from physical ability to achieve the desired motion.
11	Objective → Decision	**How should we choose the feedback law?** Introduction to optimal control; quadratic performance measures; LQR; interpretation and physical scaling of Q and R; Riccati equation computationally; comparison with pole placement; state regulation versus control effort; actuator limits and the limitations of unconstrained LQR.
12	Representation / Decision	**What can we infer from measurements?** Observability; measurement selection; state estimation; Luenberger observers; estimation-error dynamics; observer pole selection; speed versus noise tradeoffs.
13	Decision / Evaluate	**How do estimation and control work together?** Observer-based feedback; separation principle; implementation of estimated-state feedback; sensing limitations; practical estimator behavior; introduction to the Kalman-filter viewpoint.
14	Evaluate / Refine	**Will the controller work on the real system?** Validation on nonlinear models; actuator saturation; neglected dynamics; sampling; delay; quantization; measurement noise; uncertainty; stress testing; experiment; safety margins; engineering judgment; professional responsibility when analysis reveals unacceptable risk or limitations; engineering-ethics assignment.
15	Complete Design Cycle	**Putting the pieces together.** Integrated control-design case study; representation choices; behavior analysis; performance objectives; controller selection; simulation and experimental validation; comparison of design alternatives; refinement; cumulative review of the complete engineering control-design process.

The final examination period is December 14–18, 2026.

Important Fall 2026 Calendar Notes

August 31	Fall 2026 term begins.
September 7	Labor Day; no Monday class.
October 27	Last day to withdraw from Fall 2026 classes.
November 24	Modified schedule day; Thursday classes meet.
November 26–27	Thanksgiving holiday.
December 10–11	Study period; no classes.
December 14–18	Final classes / critiques / examinations.
December 18	End of Fall 2026 term.

A Final Note

Control engineering is not a collection of plots and formulas. It is a way of reasoning about dynamical systems:

What is happening?
Why is it happening?
What behavior do we want instead?
What intervention will produce it—and how do we know?

This syllabus describes the intended Fall 2026 course and may be adjusted when necessary. Material changes to grading, deadlines, or course requirements will be communicated clearly.